PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM1 | W2-PM5 | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B084J (Networkwalks)
Date	13 November 2026
Modules completed	W2-PM1 (Footprinting with Multiple Kali Tools) W2-PM5 (Zenmap Network Scanning)
Client/Target	1. Networkwalks: networkwalks.com (target assigned by the course) 2. My own local LAN network
Permission secured from client?	Yes. Permission for networkwalks.com was already secured by Networkwalks as part of the course; the local network is my own.
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had permission or that I own myself. All these materials are for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers two Week 2 modules of my Cybersecurity & Ethical Hacking course at Networkwalks. W2-PM1 footprinted the networkwalks.com domain using six built-in Kali Linux tools. W2-PM5 scanned my own local network with Zenmap to discover live hosts.

Together these modules show how an attacker moves from gathering public information about a domain to mapping live hosts on a network. Every step below includes the exact command or action used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for the reconnaissance and scanning activities.
whois	Find domain registration details (registrar, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall (WAF) protects the site.
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses, and draw a topology map.
Windows CMD (`ipconfig`)	Find my PC's local IP address, subnet mask and MAC address.

4. Activities Performed

4.1 Footprinting & Reconnaissance with Kali Tools (W2-PM1)

Footprinting means collecting public information about a target before touching it. I ran six Kali Linux tools against `networkwalks.com`, each revealing a different piece of the picture.

Tool / Command	What I found	Why it matters to an attacker
Task 1: `whois networkwalks.com`	Registrar: GoDaddy.com, LLC. Created 6 Nov 2019, expires 6 Nov 2027. Name servers: NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM. DNSSEC: unsigned.	Shows who hosts the site (HostGator). Dates and abuse contacts can help with social engineering.
Task 2: `whatweb networkwalks.com`	The site redirects HTTP to HTTPS. Apache web server, WordPress 7.1.4, WP Download Manager 3.3.76, jQuery 3.7.1. IP address 192.232.216.135. Contact email info@networkwalks.com exposed.	Exact software versions can be looked up in vulnerability databases. It also leaks the server IP and an email address.
Task 3: `nslookup networkwalks.com`	Non-authoritative answer from my local DNS server (172.20.10.1): networkwalks.com resolves to 192.232.216.135.	Knowing the real IP lets an attacker scan the server directly and map the infrastructure.
Task 4: `curl -I https://networkwalks.com`	HTTP/2 200 response. Header `server: Apache`, `x-nginx-cache: WordPress`, a session cookie (`__wpdm_client`, marked Secure and HttpOnly), and a `link` header exposing the WordPress REST API at `/wp-json/`.	Headers reveal the web server, caching setup and hidden endpoints without loading the page.

Tool / Command	What I found	Why it matters to an attacker
Task 5: `wafw00f networkwalks.com`	The site is behind a **ModSecurity (SpiderLabs)** WAF. Number of requests: 2.	Tells an attacker that simple attacks will be blocked or logged, so they must adapt.
Task 6: `dnsrecon -d networkwalks.com`	SOA and NS records (ns6135.hostgator.com 50.87.144.87, ns6136.hostgator.com 192.232.216.131), MX mail.networkwalks.com (192.232.216.135), A record 192.232.216.135, and 8 cPanel _autodiscover SRV records (cpanelemaildiscovery.net). The DNSSEC query returned no answer.	Maps the DNS footprint: name servers, mail server and the cPanel hosting setup.

Key terms in plain language:

- **WAF (Web Application Firewall):** a filter in front of a website that blocks suspicious web requests.
- **DNS:** the internet's phone book, which turns names like networkwalks.com into IP addresses.
- **NS / MX records:** NS says which servers answer DNS questions for the domain; MX says which servers receive its email.
- **SRV records:** DNS records that point to specific services (here, cPanel email auto-discovery).
- **Fingerprinting:** working out what software a system runs from the clues it gives away.

4.2 Network Scanning with Zenmap (W2-PM5)

Zenmap is the official graphical interface for Nmap, a free and open-source security scanner. I completed the seven tasks of the practical on my own network as follows.

Task	What was required	What I did / result
Task 1	Download and install Zenmap	Downloaded the Nmap installer from nmap.org/download.html and installed it with all default components (this includes Zenmap and Npcap). Version: Nmap 7.993.
Task 2	Find my local IP address and LAN subnet	Ran ipconfig in Windows CMD. IPv4 address: 172.20.10.6. Subnet mask: 255.255.255.240. LAN subnet: 172.20.10.0/28 (default gateway 172.20.10.1).
Task 3	Find live hosts in the subnet	Entered 172.20.10.0/28 in the Zenmap Target box, selected the Ping scan profile (nmap -sn 172.20.10.0/28) and clicked Scan.
Task 4	How many hosts are live?	3 hosts are live (including my own PC). Nmap scanned 16 IP addresses in 3.16 seconds.
Task 5	IP addresses of the live hosts	172.20.10.1, 172.20.10.2 and 172.20.10.6.

Task	What was required	What I did / result
Task 6	MAC addresses of the live hosts	172.20.10.1: EA:78:65:37:2C:64. 172.20.10.2: 22:74:C9:6B:0E:E5. 172.20.10.6 (my PC, from ipconfig /all): 5C-BA-EF-73-B3-AF.
Task 7	Save the topology as PDF	Opened the **Topology** tab, turned on the legend, and used **Save Graphic** to save the topology as a PDF on my desktop. File name: Nmap - Zenmap GUI.pdf

Live hosts discovered:

#	IP Address	MAC Address	Notes (device type, vendor)
1	172.20.10.1	EA:78:65:37:2C:64	Default gateway, DHCP and DNS server (router or hotspot). Vendor unknown.
2	172.20.10.2	22:74:C9:6B:0E:E5	Another device on the LAN. Vendor unknown.
3	172.20.10.6	5C-BA-EF-73-B3-AF	My own PC (Realtek RTL8821CE Wi-Fi adapter).

Reading the topology: each dot in the topology is a host, and the legend explains the colours (for example, green means a host with fewer than 3 open ports, which is expected for a ping scan that does not scan ports). In my topology, my machine (localhost) sits in the centre and the live hosts 172.20.10.1, 172.20.10.2 and 172.20.10.6 are drawn as green dots. The map also lists 192.168.56.1 and 192.168.100.100, which are not in the Nmap output of this scan; they most likely come from other scans or virtual adapters kept in the same Zenmap session, so they are not counted as live hosts of this subnet.
Note: a ping scan (`-sn`) only finds which hosts are up. It does not scan ports or services, so no vulnerabilities were tested.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers may use exposed version information to look for known weaknesses that need further review	**Medium**
2	Server IP address identifiable	Nslookup resolved the domain to `192.232.216.135`	Provides information about the network location of the web service	**Low**
3	HTTP technical information exposed	Curl returned headers showing Apache and the `/wp-json/` REST API endpoint	May assist technology fingerprinting and further enumeration	**Low**

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	**Low**
5	DNS infrastructure information exposed	DNSRecon returned SOA, NS, MX, A and SRV records (cPanel autodiscover)	DNS information helps build a broader infrastructure profile	**Medium**
6	Contact email address exposed	WhatWeb identified info@networkwalks.com	Could be used for phishing or social engineering	**Medium**
7	Multiple live hosts visible on local network	Zenmap Ping scan found 3 live hosts	Unknown or unauthorised devices may be present on a network	**Medium**

Risk level key: Critical, Medium, Low
The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering, OSINT and host discovery. No exploitation or vulnerability validation was performed as part of these four modules.

Therefore, the presence of information such as a software version, IP address, DNS record or a publicly indexed device does not by itself mean that the system is vulnerable or compromised. Further authorised security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**. Regularly review what web technologies, CMS and plugin versions are publicly visible.
2. **Keep software updated**. Update the CMS, plugins and other web technologies, and check them against current security advisories.
3. **Review HTTP headers**. Remove or reduce unnecessary technical information in HTTP response headers and consider restricting the REST API where it is not needed.
4. **Review DNS records regularly**. Check DNS records periodically so that only required information and services are public, and keep DNS software updated.
5. **Configure and monitor the WAF**. Keep the WAF (ModSecurity) enabled and tuned, since it already blocks simple attacks.
6. **Reduce public email exposure**. Avoid publishing direct contact emails where possible, or use monitored aliases to reduce phishing risk.
7. **Perform regular internal network discovery**. Scan your own network periodically to identify active devices.
8. **Investigate unknown devices**. Any unexpected device found during scanning should be identified and verified, or removed.
9. **Maintain network documentation**. Keep the network topology and device list documented and up to date.
10. **Perform security testing with authorisation**. Reconnaissance and scanning should only be done on systems where permission has been given or that you own.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking course, I completed practical activities covering footprinting with command-line tools and network scanning.

In the Kali tools activity, I learned how whois shows domain registration details, whatweb identifies web technologies, nslookup resolves domain names, curl -I shows HTTP headers, wafw00f identifies a WAF, and dnsrecon lists DNS records.

In the network scanning activity, I used Zenmap to identify my local network configuration, discover active hosts, collect IP and MAC address information, and save a network topology.

Across both modules, the common lesson was that reconnaissance is powerful precisely because it is passive: none of these tools attacked a target, yet together they built a detailed picture of it. I also learned that a good report explains what was done, what was found, what it means, what risk it may create, and how to reduce that risk. Finally, reconnaissance and scanning must always stay within an authorised scope. These activities were completed as part of the assigned educational cybersecurity course.

8. Evidences Collected

W2-PM1: Footprinting with Kali Tools

Figure 1: whois

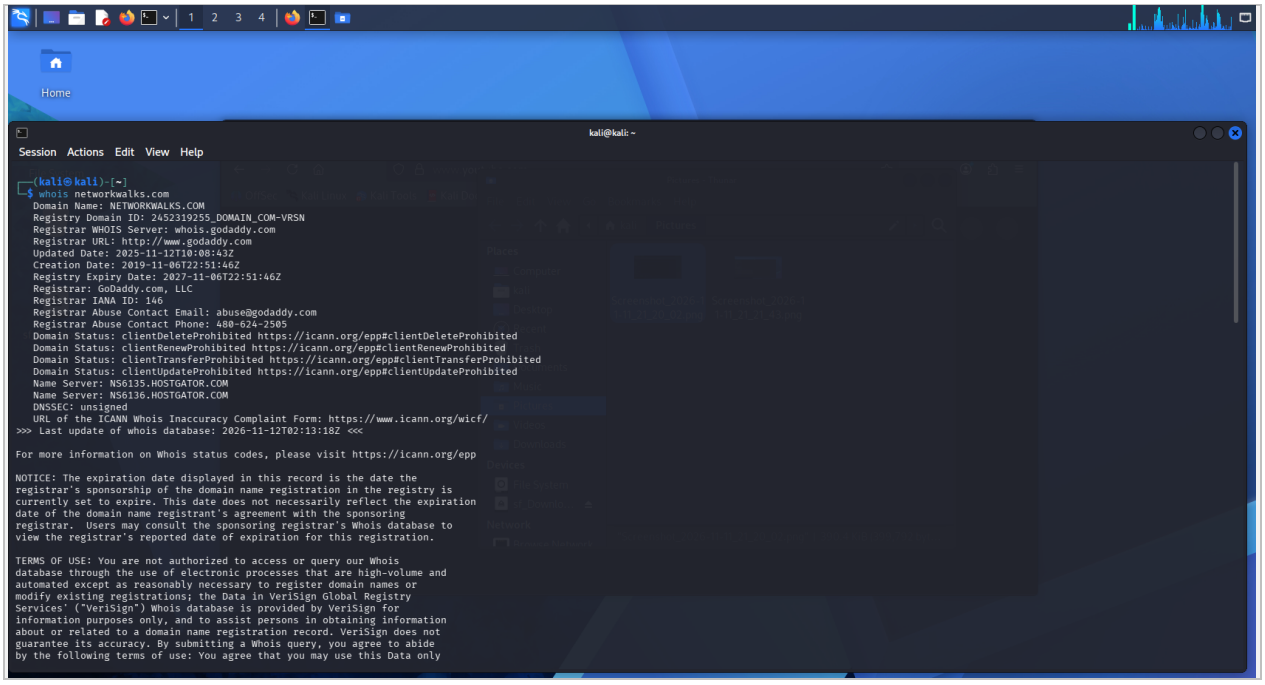


Figure 2: whatweb

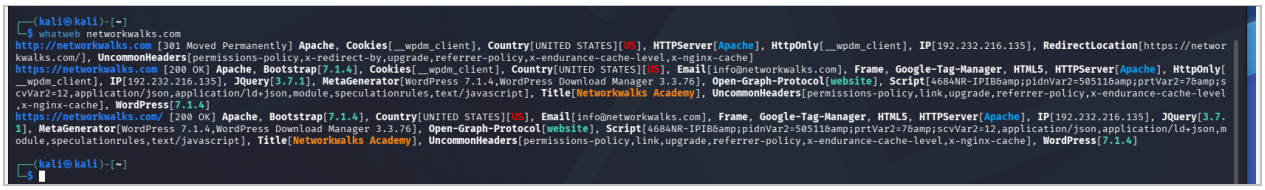


Figure 3: nslookup

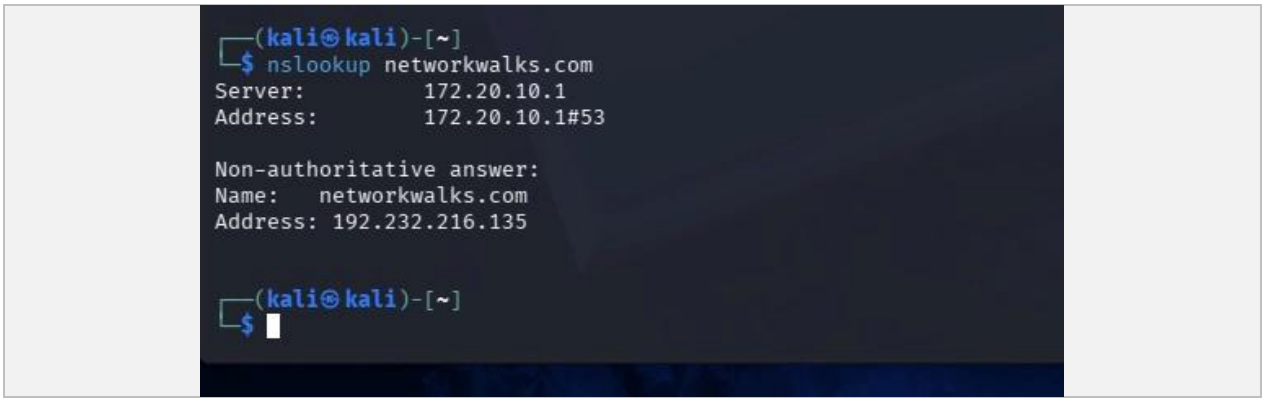


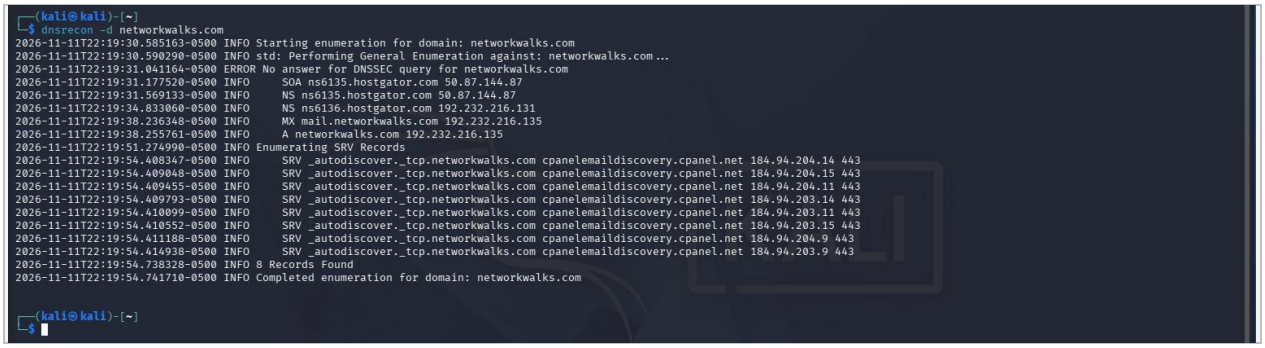
Figure 4: curl -I



Figure 5: wafw00f



Figure 6: dnsrecon



W2-PM5: Zenmap Scanning

Figure 7: Zenmap installed

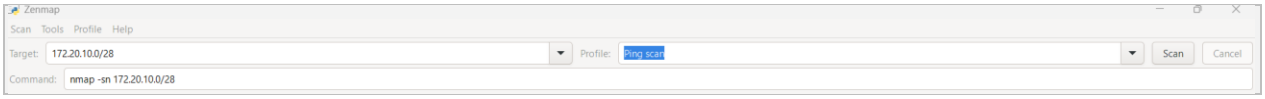


Figure 8: Local IP and subnet

```
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b0df:6b0c:6f2d:52e6%7
   IPv4 Address. . . . . . . . . . . : 172.20.10.6
   Subnet Mask . . . . . . . . . . . : 255.255.255.240
   Default Gateway . . . . . . . . . : 172.20.10.1
```

Figure 9: Ping scan results

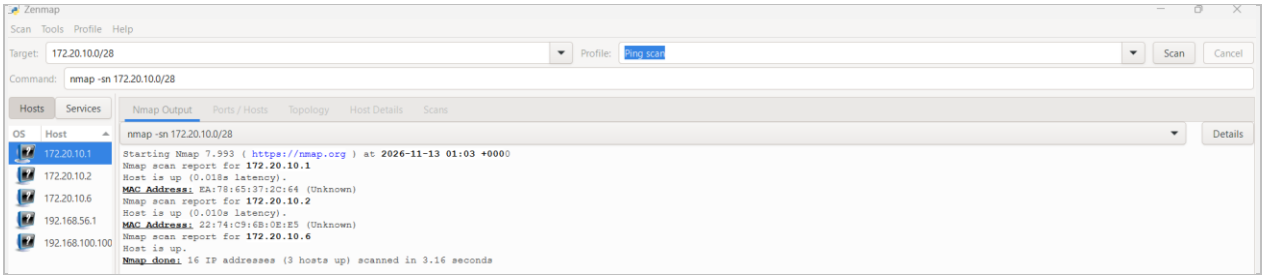


Figure 10: Local MAC address

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8821CE 802.11ac PCIe Adapter
   Physical Address. . . . . . . . . : 5C-BA-EF-73-B3-AF
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b0df:6b0c:6f2d:52e6%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.20.10.6(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.240
   Lease Obtained. . . . . . . . . . : Thursday, 12 November 2026 11:01:26 pm
   Lease Expires . . . . . . . . . . : Friday, 13 November 2026 1:31:25 am
   Default Gateway . . . . . . . . . : 172.20.10.1
   DHCP Server . . . . . . . . . . . : 172.20.10.1
   DHCPv6 IAID . . . . . . . . . . . : 308067055
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-27-0F-DC-A5-F8-0D-AC-B8-89-0C
   DNS Servers . . . . . . . . . . . : fe80::e878:65ff:fe37:2c64%7
                                       172.20.10.1
                                       fe80::e878:65ff:fe37:2c64%7
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Figure 11: Network topology

